

Load-Balanced and Congestion-Aware Routing for LEO Laser Satellite Networks Based on Deep Reinforcement Learning

Haoxin Li,¹ Junling Yuan,^{1,*} Xuhong Li,² and Qikun Zhang¹

¹Zhengzhou University of Light Industry, Zhengzhou, China

²Zhongyuan University of Technology, Zhengzhou, China

*yuanjunling@foxmail.com

Abstract—To address load imbalance in low earth orbit (LEO) laser satellite networks (LSN), this paper proposes a deep reinforcement learning (DRL) based routing algorithm, which combines proximal policy optimization (PPO) and K-shortest path (KSP) strategies to transform the large-scale routing problem into a decision-making process over a small set of paths. Simulation results demonstrate that, compared with traditional Dijkstra and random routing algorithms, the proposed algorithm fully exploits network resources, effectively prevents network bottlenecks, and significantly enhances the network's service-carrying capacity.

Keywords—LEO satellite network, laser inter-satellite link, load balancing, deep reinforcement learning, 6G

I. INTRODUCTION

With the evolution of the Internet of Things and sixth-generation (6G) technology, future networks must support ultra-low latency, high capacity, and global coverage [1]–[3]. Low earth orbit (LEO) satellite constellations, particularly LEO laser satellite networks (LSNs) enabled by laser inter-satellite links (LISLs), can provide high-bandwidth, low-interference communications that are not subject to spectrum constraints [4]. However, uneven spatiotemporal service distribution can lead to load imbalance and link congestion, thereby limiting network performance.

Traditional methods improve performance through service-aware mechanisms such as dynamic link assignment, time-scale load balancing, and backpressure routing [5]–[7], as well as congestion-aware strategies [8], but they lack adaptability. DRL-based methods, such as DDPG [9], SAC [10], and DQN [11], can achieve adaptive routing, but suffer from issues such as an overly large action space, low decision accuracy, or training instability.

To address these issues, this paper proposes an algorithm called KSP Enhanced PPO for Load-Balancing and Congestion-Aware Routing (PPO-KSP), which uses KSP to generate candidate paths, thereby reducing the action space, while also leveraging PPO to learn adaptive routing strategies under dynamic network conditions. In this way, the proposed method improves service capacity, achieves effective load balancing as well as congestion awareness.

II. SYSTEM MODEL

A. Network Model

This paper considers an LSN based on the Walker

constellation, consisting of N orbits, each with M satellites. Each satellite is connected to four neighboring satellites via LISLs, thereby forming intra-orbit and inter-orbit links. The network is modeled as a weighted graph $G=(V, L)$.

B. Problem Analysis

In LSN, each LISL $l \in L$ is allocated a bandwidth of b_l . Therefore, the total network bandwidth can be expressed as:

$$B_{\text{LSN}} = 2 \times b_l \times N \times M \quad (1)$$

The system calculates a routing path L_d for any service $d \in D$. A service will only be accepted if all links along the selected path meet the bandwidth constraints. Thus, the total number of services carried by the network is:

$$D_{\text{Num}} = \sum_{d \in D} \left(\prod_{l_d \in L_d} \alpha_{l_d} \right) \quad (2)$$

$$\alpha_{l_d} = \begin{cases} 1, & \text{if } b_d \leq c_{l_d} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Where c_{l_d} denotes the remaining available bandwidth on link l_d , b_d denotes the bandwidth required for service demands.

The average link utilization (ALU) is defined as the ratio of the link resources occupied by D_{Num} to the total resources of the entire network. It is used to reflect network resource utilization and expressed as follows:

$$\eta = \frac{\sum_{d \in D} \sum_{l_d \in L_d} b_d}{B_{\text{LSN}}} \quad (4)$$

To address the issues of underutilized resources and limited service capacity caused by load imbalance in satellite networks, this paper proposes the following optimization objective: Maximize the number of services the network can support while ensuring that no service is blocked, that is:

$$\max D_{\text{Num}}, s.t. \eta \leq 1 \quad (5)$$

III. THE PROPOSED ALGORITHM

A. DRL Model

1) *State Space*: In this paper, the agent pre-stores K candidate paths calculated by the KSP algorithm for each source-destination pair of V_s and V_d . For V_s and V_d , the

This work is supported by Henan Science and Technology Department Project No. 262300420305, 252102211112, and 252102211089.

corresponding set of candidate paths $\mathcal{P}_{V_s, V_d}$ is defined as follows:

$$\mathcal{P}_{V_s, V_d} = \{P_{V_s, V_d}^1, P_{V_s, V_d}^2, \dots, P_{V_s, V_d}^K\} \quad (6)$$

At each decision point, the state of the environment is composed of the joint features of the K candidate paths corresponding to the current service. For the P_{V_s, V_d}^k , its feature vector $f_k \in \mathbb{R}^5$ contains the following components:

$$f_k = \{c_{\min, k}, \bar{c}_k, \mu_{\max, k}, b_d^*, \text{Len}_k\} \quad (7)$$

where $c_{\min, k}$ denotes the minimum remaining bandwidth of all links on P_{V_s, V_d}^k , $\bar{c}_k$ denotes the average remaining available bandwidth of P_{V_s, V_d}^k , $\mu_{\max, k}$ denotes the maximum link utilization of P_{V_s, V_d}^k , b_d^* denotes the demand value of the current service normalized by the rated bandwidth of the network links, and Len_k denotes the path length of P_{V_s, V_d}^k .

The state space $\mathcal{S}$ observed by the agent at the time of routing decision-making can be represented as a set of feature vectors f_k , i.e.:

$$\mathcal{S} = [f_1, f_2, \dots, f_K]^T \quad (8)$$

2) *Action Space*: The action space is:

$$\mathcal{A} = \{1, 2, \dots, K\} \quad (9)$$

Each action a selects the P_{V_s, V_d}^k from the cache list as the routing path for the current service.

3) *Reward*: After the service has been successfully routed along the path selected by the agent, the reward for this action a is calculated as follows:

$$r = R + Y(a) + Z(a) - H(a) \quad (10)$$

To prevent differences in units across metrics from interfering with policy learning, the metrics are normalized in the reward function and a composite reward function is constructed using a linear weighting method. R represents the fixed positive reward obtained for successfully routing a service, $Y(a)$ is the reward term calculated based on the normalized average link utilization of the path selected by action a , $Z(a)$ is the reward term calculated based on the normalized maximum link utilization of the selected path, $H(a)$ is the penalty term calculated based on the normalized path length of the selected path. If the routing fails, the reward is set to 0.

B. Algorithm Flow

The PPO-KSP algorithm integrates the PPO algorithm from DRL with the KSP algorithm from graph theory, the latter being a classic method for computing multiple candidate paths between nodes [12]. KSP reduces the complexity of the action space, while PPO enables the learning of optimal path selection under dynamic conditions. The agent interacts with the environment, collects samples, and updates its policy using truncated objective optimization with generalized advantage estimation (GAE), thereby ensuring stable convergence [13]. The overall procedure of

the proposed PPO-KSP algorithm is summarized in Algorithm 1. (For brevity, the detailed environment interaction and constraint checking processes have been omitted.)

Algorithm 1: PPO-KSP

```

Compute KSP candidate paths  $\mathcal{P}_{V_s, V_d}$ 
Initialize  $\pi_\theta$ ,  $V_\phi$ , replay buffer  $Rp$ 
for episode=1 to  $M$  do
  Reset environment
  for  $t=1$  to  $T_{\max}$  do
    Construct state  $S_t$ 
    Select  $a_t \sim \pi_\theta(\cdot | S_t)$ , execute and obtain  $r$  via (10)
    Store  $(S_t, a_t, r)$  in  $Rp$ 
  end for
  Compute  $\hat{A}_t$ ,  $\hat{R}_t$  (GAE), update  $\pi_\theta$ ,  $V_\phi$ 
end for

```

IV. SIMULATION SETTINGS AND RESULT ANALYSIS

A. Simulation Settings

This paper presents a Python-based simulation platform for the Walker Delta satellite constellation. The constellation consists of 42 satellites distributed across 6 orbital planes, with 7 satellites in each plane. The DRL model is implemented using PyTorch, and the proposed PPO-KSP algorithm is systematically evaluated. In the comparative experiments, to simulate dynamic arrivals, the traffic follows a Poisson distribution with a density range of 900 to 2800, in increments of 100. To ensure statistical reliability, each algorithm undergoes 10 independent simulations at each density level, processing 100,000 traffic instances per simulation.

B. Simulation Result Analysis

As shown in Fig. 1, the value of K has a significant impact on both convergence behavior and final performance. When $K=20$, the algorithm performs best, indicating an appropriate balance between action diversity and training stability. Smaller values of K limit the set of candidate paths, thereby restricting the policy's expressive power and leading to less than optimal solutions. Conversely, larger values of K expand the action space, which increases the variance of the policy gradient estimates and slows down the convergence rate.

This paper uses the blocking probability (BP) to measure the network service capacity of different algorithms. The

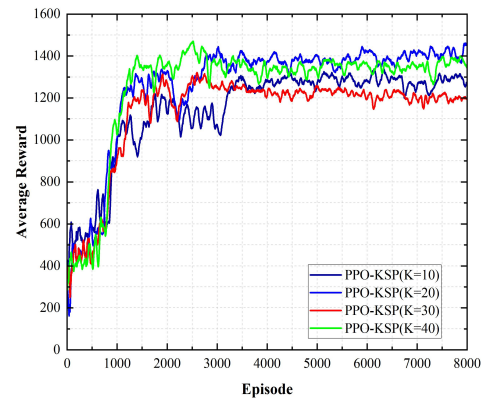


Fig. 1. Learning curves of the PPO-KSP algorithm with different K values.

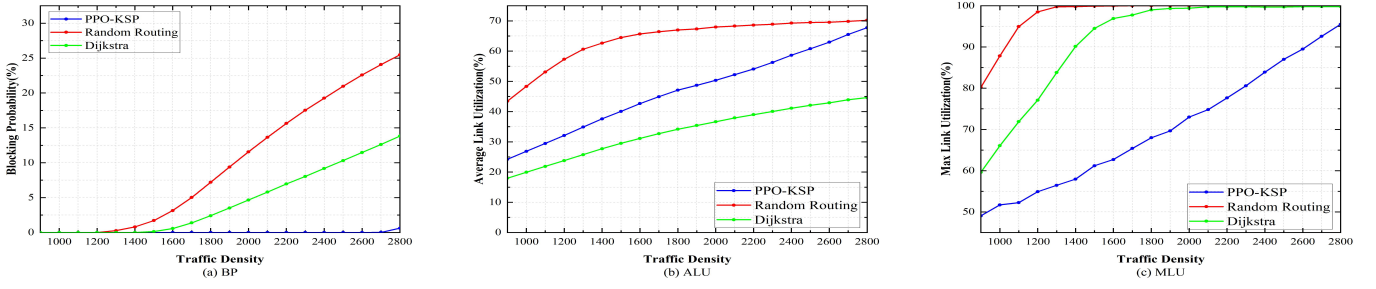


Fig. 2. Performance comparison of three routing algorithms under different traffic densities

calculation of BP is shown in (11) [14]:

$$BP = \frac{|D| - D_{Num}}{|D|} \quad (11)$$

This paper employs maximum link utilization (MLU) to evaluate the network bottleneck suppression capabilities of different algorithms. The MLU calculation is shown in (12):

$$MLU = \max_{l_i \in L} \frac{b_{l_i} - c_{l_i}}{b_{l_i}} \quad (12)$$

Fig. 2 compares the performance of different routing algorithms under various traffic densities, with Fig. 2(a), Fig. 2(b), and Fig. 2(c) showing the BP, ALU, and MLU, respectively.

As shown in Fig. 2(a), the PPO-KSP algorithm achieves the lowest BP across all traffic densities, indicating that it significantly enhances network service capacity. This improvement can be explained by the utilization characteristics observed in Fig. 2(b) and Fig. 2(c).

Specifically, Fig. 2(b) shows that PPO-KSP maintains a balanced ALU, demonstrating its ability to utilize network bandwidth resources efficiently and uniformly. Compared to the Dijkstra algorithm, which tends to concentrate traffic on the shortest path and results in underutilization of other links, PPO-KSP achieves more comprehensive utilization of overall network resources. At the same time, compared to random routing algorithms that distribute traffic randomly, PPO-KSP avoids unbalanced resource allocation and achieves more coordinated link utilization.

Fig. 2(c) further demonstrates that PPO-KSP effectively suppresses excessive growth in link utilization, thereby delaying link saturation. In contrast, the random routing algorithm often leads to early congestion due to uncontrolled traffic distribution, while the Dijkstra algorithm causes local overload by concentrating traffic on a small set of shortest paths.

V. CONCLUSION

In summary, by combining KSP-based action space reduction with PPO-based adaptive decision-making, the PPO-KSP algorithm achieves load balancing and congestion awareness. Simulation results show that, compared to traditional routing strategies, this algorithm significantly increases network service capacity by balanced resource

allocation and suppressing bottlenecks.

REFERENCES

- [1] N. Cheng, H. E. Jingchao, Z. Yin, C. Zhou, W. U. Huaqing, F. Lyu, H. Zhou, X. Shen, "6G service-oriented space-air-ground integrated network: A survey," *Chinese Journal of Aeronautics*, no. 9, pp. 1–18, 2022.
- [2] H. Guo, J. Li, J. Liu, N. Tian and N. Kato, "A survey on space-air-ground-sea integrated network security in 6G," in *IEEE Communications Surveys & Tutorials*, vol. 24, no. 1, pp. 53–87, First quarter 2022.
- [3] M. Y. Abdelsadek, et al., "Future space networks: Toward the next giant leap for humankind," in *IEEE Transactions on Communications*, vol. 71, no. 2, pp. 949–1007, Feb. 2023.
- [4] R. Li, B. Lin, Y. Liu, M. Dong and S. Zhao, "A survey on laser space network: Terminals, links, and architectures," in *IEEE Access*, vol. 10, pp. 34815–34834, 2022.
- [5] X. Qi, et al., "A dynamic link allocation strategy for efficient load balancing in multi-layer satellite networks," *2025 IEEE International Conference on Communications Workshops (ICC Workshops)*, Montreal, QC, Canada, 2025, pp. 80–85.
- [6] S. Liu, J. Liu and B. Xia, "Adaptive timescale load balancing routing algorithm for LEO satellite network," *2023 IEEE/CIC International Conference on Communications in China (ICCC)*, Dalian, China, 2023, pp. 1–5.
- [7] W. Deng, X. Lv and J. Liu, "Delay-optimized backpressure algorithm for load balancing in LEO satellite networks," in *IEEE Communications Letters*, vol. 30, pp. 922–926, 2026.
- [8] Y. Ning, et al., "Load-balancing routing algorithms for service congestion avoidance in LEO optical satellite networks," in *Journal of Optical Communications and Networking*, vol. 15, no. 12, pp. 1038–1049, Dec. 2023.
- [9] Y.-H. Hsu, J.-I. Lee and F.-M. Xu, "A deep reinforcement learning based routing scheme for LEO satellite networks in 6G," *2023 IEEE Wireless Communications and Networking Conference (WCNC)*, Glasgow, United Kingdom, 2023, pp. 1–6.
- [10] J. Zhu, X. Sun and D. Chen, "Deep reinforcement learning for congestion control and routing optimization in LEO satellite networks," *2023 5th International Conference on Artificial Intelligence and Computer Applications (ICAICA)*, Dalian, China, 2023, pp. 739–742.
- [11] F. Dong, et al., "DRL-based load-balancing routing scheme for 6G space-air-ground integrated networks," *Remote Sensing*, no. 11, 2023.
- [12] J. Yuan, Q. Zhang, and X. Zhang, "A routing modulation and spectrum assignment algorithm for semi-filterless elastic optical networks," *Optical Switching and Networking*, vol. 51, pp. 1–9, Feb. 2024.
- [13] A. d. Rio, D. Jimenez and J. Serrano, "Comparative analysis of A3C and PPO algorithms in reinforcement learning: A survey on general environments," in *IEEE Access*, vol. 12, pp. 146795–146806, 2024.
- [14] J. Yuan, et al., "Service provisioning for anycast requests in semi-filterless optical networks with spectrum and IT resource balancing," *Optical Fiber Technology*, vol. 93, 2025.